Tutorial - 3

(2) (i)

$$A^2 \int_{-a}^{a} (a^2 - x^2)^2 \, dx = 1$$

$$\Rightarrow 2A^2 \int_0^a a^4 + x^4 - 2a^2x^2 \, dx$$

$$\Rightarrow 2A^2 \left[a^4 x + \frac{x^5}{5} - \frac{2a^2 x^3}{3} \right]_0^a = 1$$

$$\Rightarrow 2A^2 \left[a^5 + \frac{a^5}{5} - \frac{2a^5}{3} \right]$$

$$\Rightarrow 2A^2 a^5 \left(\frac{15 + 3 - 10}{15} \right) \Rightarrow \frac{16}{15} A^2 a^5 = 1$$

$$\Rightarrow A = \left(\frac{15}{16 a^5} \right)^{1/2} \text{ (Ans)}$$

(ii) $\hat{T} = \frac{\hat{p}^2}{2m} \Rightarrow \langle \hat{T} \rangle = \frac{\langle \hat{p}^2 \rangle}{2m}$

$\therefore \langle \hat{p}^2 \rangle = A^2 \int (a^2 - x^2) \left(-\hbar^2 \frac{\partial^2}{\partial x^2}\right)(a^2 - x^2)\, dx$

$= -\hbar^2 A^2 \int (a^2 - x^2)(-2)\, dx$

$= 2\hbar^2 A^2 \int_{-a}^{a} a^2 - x^2\, dx.$

$= 4\hbar^2 A^2 \left[a^2 x - \frac{x^3}{3}\right]_0^a$

$= \cancel{4}\hbar^2 \cdot \frac{\cancel{15}^{5}}{\cancel{16}_{28} a^5} \cdot \frac{\cancel{2}a^3}{\cancel{3}} = \frac{5\hbar^2}{2a^2}$

$\therefore \boxed{\langle \hat{T} \rangle = \frac{5\hbar^2}{4a^2 m}}$ (Ans)

(iii) $\langle x \rangle = A^2 \int_{-a}^{a} (a^2 - x^2)^2 x\, dx = 0$ [odd Integral]

$\therefore \langle x^2 \rangle = A^2 \int_{-a}^{a} (a^2 - x^2)^2 x^2\, dx$

$= 2A^2 \int_0^a (a^4 x^2 + x^6 - 2a^2 x^4)\, dx.$

$= 2A^2 \left[\frac{a^4 x^3}{3} + \frac{x^7}{7} - \frac{2a^2 x^5}{5}\right]_0^a$

$= 2A^2 a^7 \left[\frac{35 + 15 - 42}{105}\right]$

$= \cancel{2} \cdot \frac{\cancel{15}}{\cancel{16}_{8} a^5} \cdot a^7 \cdot \frac{\cancel{28}\,8}{\cancel{105}\,7} = \frac{\cancel{28}}{56} \frac{a^2}{7}$

$\therefore \langle p \rangle = A^2 \int (a^2 - x^2) - i\hbar \frac{\partial}{\partial x} (a^2 - x^2)\, dx.$

$= -i\hbar A^2 \int_{-a}^{a} (a^2 - x^2)(-2x)\, dx = 0$ [odd Integral].

$\therefore \langle \hat{p}^2 \rangle = \frac{5\hbar^2}{2a^2}$

$\therefore \Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{a}{\sqrt{7}}.$

$\Delta p = \sqrt{\frac{5\hbar^2}{2a^2}} = \frac{\sqrt{5}\hbar}{\sqrt{2}\, a}$

(iv) $\Delta x \Delta p = \sqrt{\frac{5}{14}}\hbar > \frac{\hbar}{2}$. Hence, Heisenberg's uncertainity principle is verified.

(1) (a) $G(\lambda) = e^{\lambda\hat{A}}\,\hat{B}\,e^{-\lambda\hat{A}}$

$$\therefore G(\lambda) = G(0) + \frac{1}{1!}\frac{dG}{d\lambda}\Big|_{\lambda=0} + \frac{1}{2!}\frac{d^2G(\lambda)}{d\lambda^2}\Big|_{\lambda=0} + \dots$$

$$\therefore \frac{dG}{d\lambda} = \frac{d}{d\lambda}\left(e^{\lambda A} B e^{-\lambda A}\right)$$

$$= (A e^{\lambda A})(B e^{-\lambda A}) - (e^{\lambda A})(B A e^{-\lambda A})$$

$$= e^{\lambda A}[AB - BA]\,e^{-\lambda A}$$

$$= e^{\lambda A}[A, B]\,e^{-\lambda A}$$

$$\therefore \frac{d^2G}{d\lambda^2} = \frac{d}{d\lambda}\left(e^{\lambda A}[A,B]e^{-\lambda A}\right)$$

$$= A e^{\lambda A}[A,B]e^{-\lambda A} - e^{\lambda A}[A,B]A e^{-\lambda A}$$

$$= e^{\lambda A}[A,[A,B]]\,e^{-\lambda A}$$

$$\therefore G(\lambda) = B + \frac{1}{1!}[A,B] + \frac{1}{2!}[A,[A,B]] + \dots$$

$$\Rightarrow \cancel{G(\lambda)} = e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \frac{1}{1!}[\hat{A},\hat{B}] + \frac{1}{2!}[\hat{A},[\hat{A},\hat{B}]] + \dots$$

(Proved) (1)

(@ $\lambda = 0$)

(b) $G(\lambda) = e^{\lambda\hat{A}}e^{\lambda\hat{B}}$

$$\therefore \frac{dG}{d\lambda} = A e^{\lambda A} e^{\lambda B} + e^{\lambda A} B e^{\lambda B}$$

$$= A G(\lambda) + e^{\lambda A} B e^{-\lambda A} G(\lambda)$$

Taking $\hat{A} \to \lambda\hat{A}$ & using the Hadamard Lemma:

$$e^{\lambda\hat{A}}\hat{B}e^{-\lambda\hat{A}} = \hat{B} + \frac{\lambda}{1!}[\hat{A},\hat{B}] + \frac{\lambda^2}{2!}[\hat{A},[\hat{A},\hat{B}]] + \dots$$

$$\therefore \frac{dG}{d\lambda} = AG + \left(B + \frac{\lambda}{1!}[A,B] + \frac{\lambda^2}{2!}[A,[A,B]] + \dots\right)G$$

$$= \left(\hat{A} + \hat{B} + \frac{\lambda}{1!}[\hat{A},\hat{B}] + \frac{\lambda^2}{2!}[\hat{A},[\hat{A},\hat{B}]] + \dots\right)G(\lambda)$$

$$\therefore [\hat{A},[\hat{A},\hat{B}]] = 0.$$

(4) (Proved)

$$\therefore \frac{dG}{d\lambda} = \left(A + B + \lambda[A,B]\right)G(\lambda).$$

$$\Rightarrow \int_1^{G} \frac{dG}{G} = \int_0^{\lambda} (A+B)\,d\lambda + [A,B]\int_0^{\lambda} \lambda\, d\lambda$$

$$\Rightarrow \ln G(\lambda) = (A+B)\lambda + \frac{\lambda^2}{2}[A,B].$$

$$\cancel{\Rightarrow G(\lambda) = e^{A}}$$

$$\Rightarrow \boxed{G(\lambda) = e^{(\hat{A}+\hat{B})\lambda + \frac{\lambda^2}{2}[\hat{A},\hat{B}]}} \quad \text{(proved)} \quad (5)$$

@ $\lambda = 1$:

$$e^{\hat{A}} e^{\hat{B}} = e^{(\hat{A}+\hat{B}) + \frac{1}{2}[\hat{A},\hat{B}]}. \quad (6)$$

If $[\hat{A},\hat{B}] = 0$: $e^{\hat{A}} e^{\hat{B}} = e^{(\hat{A}+\hat{B})}$ (7)